

Credit Card Fraud Detection using Machine Learning

Introduction

Credit card fraud has become a major issue in digital transactions. This project aims to develop a machine learning model that can identify fraudulent transactions based on transaction data.

Objective

To classify transactions as fraudulent or legitimate using machine learning algorithms.

Dataset

The project uses two datasets:

- fraudTrain.csv
- fraudTest.csv

Tools and Technologies

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Methodology

1. Data Loading
2. Data Cleaning
3. Exploratory Data Analysis
4. Feature Encoding
5. Missing Value Handling
6. Data Scaling
7. Model Training
8. Model Evaluation

Algorithms Used

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Conclusion

Among all implemented algorithms, Random Forest provided the best performance for detecting fraudulent transactions.

Future Scope

- Hyperparameter Tuning
- XGBoost Implementation
- Real-Time Fraud Detection System